

1.a) Define and explain the major functions of the Data Link Layer in handling node-to-node communication issues. (3M)

b) A sender transmits frames of size **1000 bits** over a link with a rate of **1 Mbps**. If the propagation delay is **10 ms**, calculate the **transmission time** and **total delay**. (4M)

c) Discuss how framing and flow control ensure reliable communication in real-world networks. (3M)

2.a) Define framing and explain different techniques used to divide a bit stream into data units. (3M)

b) Apply **bit stuffing** to the data: **011111101111110** using flag **01111110** and show the transmitted frame. (4M)

c) Evaluate which framing technique is suitable for applications where data frequently contains control-like patterns. (3M)

3.a) Define sliding window protocol and explain its working. (3M)

b) For a system with **window size = 4**, determine the **minimum number of sequence numbers required** for:

i) Go-Back-N

ii) Selective Repeat (4M)

c) Analyze the practical impact of Go-Back-N vs Selective Repeat in high-error networks. (3M)

4.a) Explain error detection and error correction techniques used in data communication. (3M)

b) A data word **1101011011** is transmitted using **CRC** with generator polynomial **10011**. Determine the transmitted codeword and check for errors if the received word is **11010110111010**. (4M)

c) Discuss scenarios where error correction is preferred over retransmission in modern networks. (3M)

5.a) Define CSMA/CD and explain how it works in shared networks. (3M)

b) A network has **bandwidth = 10 Mbps** and **maximum propagation delay = 25.6 μs**. Calculate the **minimum frame size** required for collision detection. (4M)

c) Analyze how propagation delay affects the efficiency of Ethernet networks. (3M)

6.a) Define ARQ protocols and explain their role in reliable data transmission. (3M)

b) A frame **1011011** is transmitted with **even parity**. If the received frame is **1011111**, determine whether an error occurred. (4M)

c) Evaluate how retransmissions affect throughput in noisy communication channels. (3M)

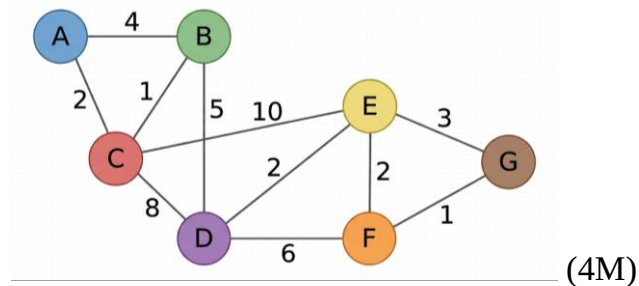
7.a) Define multiple access protocols and explain challenges in shared communication channels. (3M)

b) For **Pure ALOHA**, if the offered load $G = 0.5$, calculate the **throughput**. Also compute throughput for **Slotted ALOHA** with same G . (Use: $S = G e^{-2G}$ and $S = 2G e^{-G}$) (4M)

c) Discuss why carrier sensing improves performance in real-time communication systems. (3M)

8.a) Define shortest path routing and explain its principle. (3M)

b) Using **Dijkstra's algorithm**, find the shortest path from node **A** to **G** for the graph:



c) Analyze limitations of shortest path algorithms in dynamic networks. (3M)

9.a) Define congestion in packet-switched networks and explain its causes. (3M)

b) A router processes **1000 packets/sec**, but receives **1200 packets/sec**. If buffer size is **500 packets**, calculate how long before overflow occurs. (4M)

c) Analyze how QoS parameters influence congestion management in real-time applications. (3M)

10.a) Define leaky bucket and token bucket algorithms. (3M)

b) A token bucket has:

- Bucket size = 10 KB
- Token rate = 2 KB/sec

If 15 KB data arrives at once, determine:

- Data transmitted immediately
- Data delayed (4M)

c) Evaluate which algorithm is better for bursty traffic like video streaming. (3M)